Topic 2 – Pole Zero and Filter Response

Exercise 3.1: Consider the following filter:

$$H(z) = \frac{z^8 - 0.9z^7 - 1.375z^6 - 0.975z^5 + 4.8125z^4 - 0.975z^3 - 1.375z^2 - 0.9z^1 + 1}{z^8}$$

In MATLAB

a. Find all the poles and zeros of this filter and plot them in the z-plane

```
B=[1 -0.9 -1.375 -0.975 4.8125 -0.975 -1.375 -0.9 1]; %H(z)'s numerator coeffs
A = 1; %H(z)'s denominator coeffs
Zeros = roots(B); %Zeros of filter
zplane(Zeros,zeros(8,1))
title('filter''s z-plane')
```

b. Verify that the zeros of this filter are reciprocal to each other.

Zeros =

-1.0000 + 1.0000i
-1.0000 - 1.0000i
-0.5000 + 0.5000i
-0.5000 - 0.5000i
1.2000 + 0.4000i
1.2000 - 0.4000i
0.7500 + 0.2500i
0.7500 - 0.2500i

It can be checked that each element in 'Zeros' is the reciprocal of another i.e., Zeros(i) = 1/Zeros(k); 1<= i,k <=8

Topic 2 – Pole Zero and Filter Response

c. Plot the phase response of this filter to identify its special property

```
[H,w]=freqz(B,A);
figure
plot(unwrap(angle(H)))
```

d. Find the impulse response of this filter

```
h=impz(B,1);
```

e. The input of this filter is x[n]=[1,3,5,7,9,11,9,7,5,3,1] (n=0,1,2,…10). Find the output of the filter

```
x=[1,3,5,7,9,11,9,7,5,3,1];
y=conv(x, h);
```

Topic 3 – Magnitude and Phase Response

Exercise 3.2: Given a digital system with the following expression

$$H(z) = \frac{1-2z+3z^2}{3-2z+z^2}$$

In MATLAB

a. Compute and plot the magnitude response of the system. Identify the special property of this system.

```
B=[3, -2, 1];
A=[1, -2, 3];
[H,w]=freqz(B,A);
plot(w, abs(H));%it can be seen |H| is constant ->all-pass filter
xlabel('frequency (rad)');
ylabel('magnitude response')
axis([0 pi 0 2])
```

b. Find the zeros and poles of this system, demonstrate that the zeros are reciprocal of the poles and show their location on the z-plane

```
zeros=roots(B);
poles=roots(A);%it can be verified each pole is a reciprocal of a zero
zplane(zeros, poles);
```

c. Compute and plot the phase response of the system

```
plot(unwrap(angle(H)))
```

Topic 3 – Phase and Group Delay

Exercise 3.3: A digital filter has eight trivial poles locating at the origin and eight zeros locating at

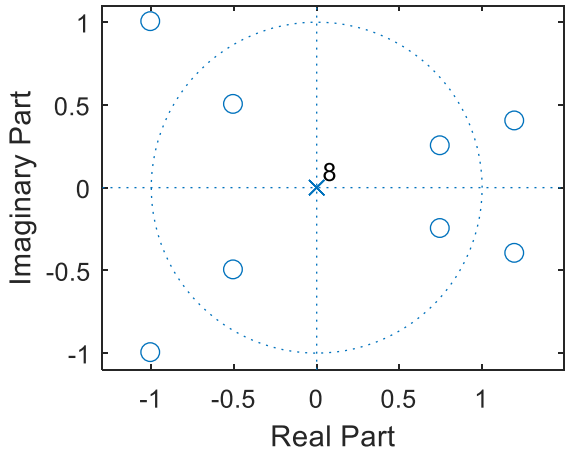
Zeros	$0.75 \pm j\,0.25$; $-0.5 \pm j\,0.5$; $1.2 \pm j\,0.4$; $-1 \pm j$

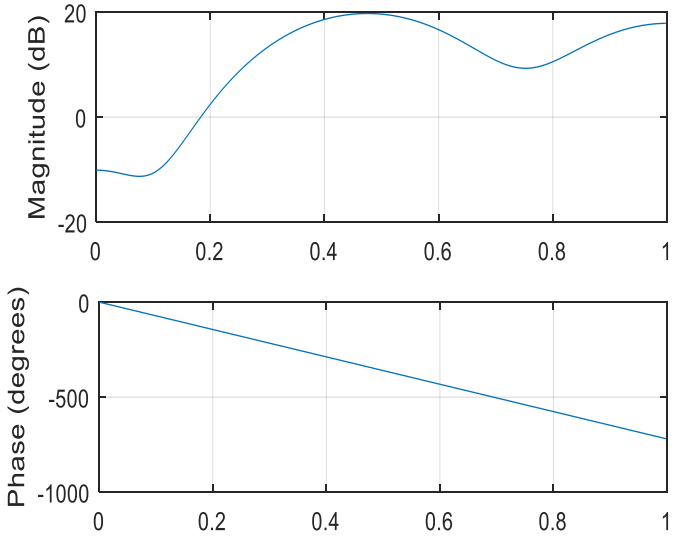
Using MATLAB,

a. Plot the pole-zero plane (z-plane) and a frequency response of this filter. Comment on its phase response and the location of zeros. -> each zero is a reciprocal of another zero, phase response is linear

```
z = [0.75+j*0.25;0.75-j*0.25; -0.5+j*0.5; -0.5-j*0.5; 1.2-j*0.4; 1.2+j*0.4; -1-j; -1+j];
B = poly(z);
zplane(z,zeros(8,1))
figure; freqz(B,1);
```

b. Calculate the group delay of this filter





Topic 3 – Min/Max Phase Filters

Exercise 3.4: The digital filters A, B, and C have four trivial poles locating at the origin and four zeros locating at

	Zeros Location
Filter A	$0.75 \pm j\,0.25$; $-0.5 \pm j\,0.5$
Filter B	$1.2 \pm j\,0.4$; $-0.5 \pm j\,0.5$
Filter C	$1.2 \pm j\,0.4$; $-1 \pm j$

Using MATLAB,

a. Plot the pole-zero plane (z-plane) and a frequency response of each filter.
b. Are there any differences in their magnitude responses?
c. Comment on the phase lag of each of the filters.
d. Classify each of the above filters as either maximum phase, mixed phase or minimum phase filters

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Topic 3 – Min/Max Phase Filters

```
z_A = [0.75+j*0.25;0.75-j*0.25; -0.5+j*0.5; -0.5-j*0.5]
z_B = [1.2+j*0.4; 1.2-j*0.4; -0.5+j*0.5; -0.5-j*0.5];
z_C = [1.2+j*0.4; 1.2-j*0.4;-1-j;-1+j];
b_A = poly(z_A); b_B = poly(z_B); b_C = poly(z_C);
[HA,w]=freqz(b_A,1); [HB,w]=freqz(b_B,1);
[HC,w]=freqz(b_C,1);
figure
subplot 331
zplane(z_A, zeros(4,1));
xlabel(''); ylabel(''); title('Filter A')
subplot 334
plot(w/w(end),20*log10(abs(HA)))
ylabel('magnitude (dB)')
grid on
axis([0 1 -20 20])
subplot 337
plot(w/w(end),unwrap(angle(HA))/pi*180)
xlabel('normalized frequency')
ylabel('phase (degree)')
axis([0 1 -800 100])
```

```
subplot 332
zplane(z_B, zeros(4,1));
xlabel(''); ylabel(''); title('Filter B')
subplot 335
plot(w/w(end),20*log10(abs(HB)))
axis([0 1 -20 20]); grid on
subplot 338
plot(w/w(end),unwrap(angle(HB))/pi*180)
xlabel('normalized frequency')
axis([0 1 -800 100])
subplot 333
zplane(z_C, zeros(4,1));
xlabel(''); ylabel(''); title('Filter C')
subplot 336
plot(w/w(end),20*log10(abs(HC)))
axis([0 1 -20 20]); grid on
subplot 339
plot(w/w(end),unwrap(angle(HC))/pi*180)
xlabel('normalized frequency')
axis([0 1 -800 100])
```

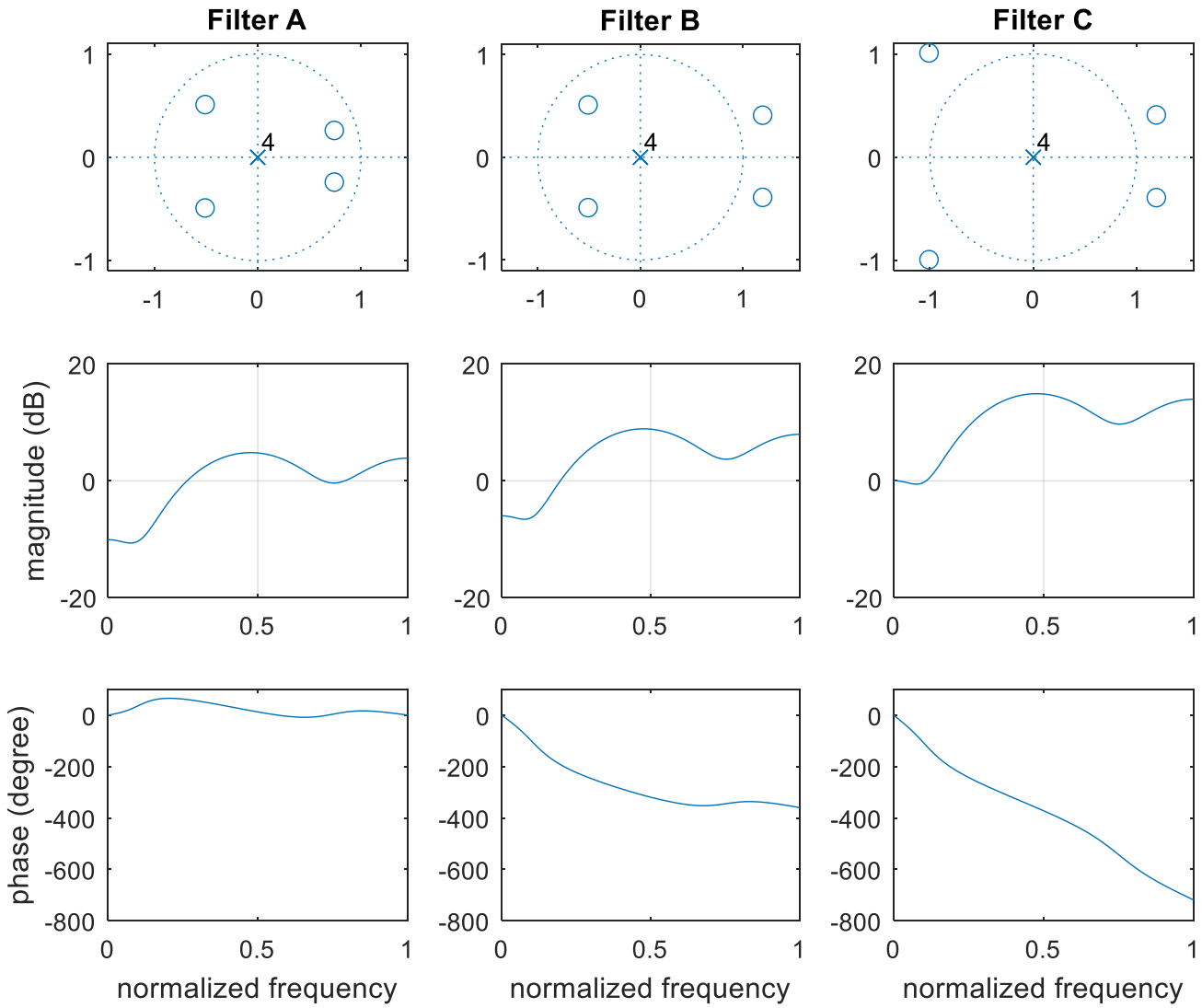

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Topic 3 – Min/Max Phase Filters

a. Are there any differences in their magnitude responses? -> variation is identical, each different with the others by a gain factor

b. Filter A, min phase lag
Filter B, medium phase lag
Filter C, max phase lag

c. Filter A, minimum phase
Filter B mixed phase
Filter C, maximum phase



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Topic 3 – Cut-off Frequency and Bandwidth

Exercise 3.5: A digital filter is given by the following transfer function

$$H(z) = \frac{0.2}{1-0.8z^{-1}}$$

1. Find the cut-off frequency and bandwidth of the filter
2. Using MATLAB,
    a. Compute and plot the magnitude response of the system.
    b. Write code segment to automatically compute the cut-off frequency and bandwidth of the filter. Compare with the result in question 1

```
B=0.2; % H(z)'s numerator coefficients
A=[1 -0.8]; % H(z)'s denominator coefficients
[H,w]=freqz(B,A,2048); % compute frequency response
Hm = 20*log10(abs(H)); % compute magnitude response in dB
wc = w(max(find(Hm>-3))); % find cut-off frequency
BW = wc-0; % bandwidth of filter
plot(w,Hm); hold on
plot(wc*ones(1,10),linspace(-20, Hm(find(w==wc)),10),'--');
axis([0, pi, -20, 0]); xlabel('frequency (rad)')
ylabel('magnitude(dB)')
```

